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ORIGINAL PAPER

Validity and responsiveness of EQ-5D-Y in children with haematological malignancies and their caregivers

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Abstract
The psychometric properties of the EQ-5D-Y have not been widely tested in severely ill children. The aim of this study was to assess and compare the validity and responsiveness of the EQ-5D-Y-3L and EQ-5D-Y-5L in paediatric inpatients with haematological malignancies and caregivers. Respondents completed the interviewer-administered self-complete or proxy version of the EQ-5D-Y-3L and EQ-5D-Y-5L and an overall health assessment twice on different days. Known-groups validity was assessed by comparing patients who differed in overall health and Eastern Cooperative Oncology Group (ECOG) performance. Responsiveness to worsened health was assessed using standardised effect size (SES) for patients with worsened ECOG grade, self-reported rating, or chemotherapy initiation. Ninety-six dyads completed the baseline questionnaires. A smaller proportion of patients reported "no problems" on the EQ-5D-Y-5L compared to EQ-5D-Y-3L for most of the five dimensions. Patients in poor health reported more problems in all dimensions and had higher EQ-5D-Y-5L level sum score, lower EQ VAS and EQ-5D-Y-3L index scores (Cohen's d ES: 0.32–1.38 for patients; 0.50–2.05 for caregivers). There was a mild to good responsiveness to worsened health condition based on ECOG (SES: 0.14–0.61 for patients; 0.40–0.96 for caregivers), suggesting the proxy version was slightly responsive than the self-complete version of both instruments. The results demonstrated validity and responsiveness for both the self-complete and proxy versions of the EQ-5D-Y-3L and EQ-5D-Y-5L. The proxy and 5-level versions of the instrument were more sensitive than the self-complete and 3-level versions in this patient group.

Keywords EQ-5D-Y-5L · EQ-5D-Y-3L · Proxy · Validity · Responsiveness · Haematological malignancies

JEL Classification I19

Introduction

The EQ-5D-3L is a widely used generic instrument to measure and value health outcomes in a wide range of patient and non-patient adult populations [1]. Given that

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some of the language and content used in the EQ-5D-3L might not be suitable for younger populations, a youth version of the instrument, the EQ-5D-Y, was developed for use in children and adolescents aged 8–17 years [2]. The EQ-5D-Y-3L (Y-3L) retains the same five-dimension, three-level format of the EQ-5D-3L, but the dimensions and levels are described in more appropriate language for children and adolescents [2, 3]. Because the 3-level EQ-5D was considered potentially insensitive to small change in health, a 5-level EQ-5D (EQ-5D-5L) [4] was developed and demonstrated to have superior measurement properties than its original [5–7]. The success of EQ-5D-5L led to the development of a 5-level EQ-5D-Y (Y-5L) [8].

Prior to using a generic instrument like EQ-5D-Y in clinical practice or research, it is necessary to test its psychometric properties, such as reliability, validity, and responsiveness. The Y-3L has been shown to be a reliable, valid, and sensitive instrument in several conditions and settings [9–14]. The Y-5L has been validated in children with different health conditions, such as idiopathic scoliosis [15], haematological diseases[16], psychological disorder[17], and general young population[18]. Moreover, a head-to-head comparison of the responsiveness of Y-3L and Y-5L in children and adolescents receiving acute orthopaedic care was conducted [19]. To date, however, the self-complete versions of the Y-3L or Y-5L have not been widely tested or compared in paediatric cancer patients.

There are proxy versions of Y-3L and Y-5L designed for parents or other caregivers to assess children's health they take care of. Proxy reports can complement children's perception of their own health and are useful when children are not able to evaluate themselves due to poor literacy or health status [12, 20–22]. The proxy version of the Y-3L has demonstrated validity for assessing children over 6 years of age [20] and was found to be more reliable than the self-report version in a Japanese study [23]. The proxy version of Y-5L demonstrated better responsiveness than the Y-3L proxy version in a study of children with acute lymphoblastic leukaemia [24].

The aim of this study was to simultaneously assess the validity and responsiveness of the EQ-5D-Y (Y-3L) and the recently developed EQ-5D-Y-5L (Y-5L) in Chinese children and adolescents with haematological malignancies. Haematological malignancies are a leading cause of morbidity and mortality in paediatric populations, with health related quality of life (HRQoL) being affected not only by the disease itself but also by the side effects of radiation and chemotherapy [25]. Since children and adolescents with haematological malignancies may be too ill to report their own HRQoL, we evaluated both the self-complete and proxy versions of the two EQ-5D-Y questionnaires in this study.

Methods

Sampling

Paediatric inpatients with leukaemia or other haematological malignancies and their caregivers were recruited from the Shanghai Children's Medical Centre between November 2018 and September 2019. The inclusion criteria for patients were: (1) a diagnosis of leukaemia or other haematological malignancy; (2) aged 8–17 years; (3) hospitalisation for chemotherapy; and (4) ability to converse in Chinese and understand the study questionnaire. Children who were not well enough to be interviewed or able to cooperate due to cognitive impairment or mental disorders did not give assent or whose legal guardians did not give consent were excluded. The inclusion criteria for caregivers were: (1) adult family member of an eligible patient; (2) taking care of the patient on the day of survey; (3) ability to converse in Chinese; and (4) informed consent. Caregivers who were cognitively unwell or unwilling to take part were excluded.

Procedures

All consenting patients and their caregivers were face-to-face interviewed in the haematology wards by a trained clinical pharmacist. The self-complete and proxy 1 versions of EQ-5D-Y were administrated. In all cases, the interviewer read out loud all the questions and response options. All participants were interviewed again in the same wards 2–13 days after their first interviews. For each patient–caregiver dyad, the caregiver was interviewed first. The baseline questionnaire for caregivers included: (1) questions on the patient's birth date, sex, disease duration, and level of education; (2) the proxy version of the Y-5L questionnaire; (3) the proxy version of an overall health assessment (OHA) question; (4) questions on the caregiver's relationship with the patient, age, educational attainment, monthly household income, and residential area; and (5) the proxy version of the Y-3L questionnaire (without the EQ VAS). The baseline questionnaire for patients included: (1) the Y-5L questionnaire; (2) the OHA question; and (3) the Y-3L questionnaire (without the EQ VAS). The structure of the follow-up interviews was the same as the baseline interviews except that the order of the Y-5L and Y-3L questionnaires was swapped and the demographical questions were omitted.

On the days of the baseline and follow-up interviews, the patients were assessed by the interviewer using the Eastern Cooperative Oncology Group (ECOG) performance scale[26] and their clinical diagnoses and chemotherapy treatment status (initiated/not initiated) were ascertained from patients' medical records.

Study instruments

The Y-3L questionnaire includes a five-dimension health-status descriptive system and a visual analogue scale (EQ VAS). The five dimensions forming the descriptive system are: mobility/walking about, looking after myself, doing usual activities, having pain or discomfort, and feeling worried, sad or unhappy. Both Y-3L and Y-5L include the same dimensions, but each dimension in the Y-3L has three response options corresponding to the severity levels of no problems, some problems, and a lot of problems, while the Y-5L includes five levels of severity for each dimension, corresponding to no problems, a little bit of a problem, some problems, a lot of problems, and extreme problems/unable [8]. The EQ VAS is a vertical, hash-marked visual analogue scale anchored by 0 (the worst imaginable health) at the bottom and 100 (the best imaginable health) at the top for respondents to assess their overall health on the day of the survey. The proxy versions of the Y-3L and Y-5L are identical to their respective self-complete versions except that the response options are worded in the third person, for example, "He/she has no problem walking about". There are two proxy versions for the two EQ-5D-Y instruments. Proxy version 1 asks the proxy to provide their own impression of the patient's health on the day of the interview; proxy version 2 asks the proxy to try to imagine how the patient would rate his/her own health [27]. We elected to test proxy version 1 in this study.

The overall health assessment (OHA) question has been shown to be a validated measure of subjective health in children and adolescents [28]. Moreover, it served as an outcome measure for testing the validity and reliability of the EQ-5D-Y-3L in multinational studies [19, 29]. The question in the present study was phrased as "How is your overall health today? Is it excellent, good, fair, poor or very poor"? Its proxy version was phrased for respondents to assess their children's overall health on the day of survey.

The Eastern Cooperative Oncology Group (ECOG) performance scale classifies patients into one of five grades: 0 (fully active, no performance restrictions); 1 (restricted in physically strenuous activity but ambulatory, able to carry out light or sedentary work); 2 (ambulatory and capable of all self-care but unable to carry out any work activities, up and about more than 50% of waking hours); 3 (capable of only limited self-care, confined to bed or chair more than 50% of waking hours); 4 (completely disabled; cannot carry out any self-care, totally confined to bed or chair); and 5 (dead) [26].

Data analysis

The self-complete and proxy versions of Y-3L and Y-5L were assessed separately for the distributional characteristics, construct validity, and responsiveness of their dimensional items as well as the Y-3L index scores, Y-5L level sum scores (LSS), and EQ VAS.

Known-groups validity was assessed by comparing patients who were in 'good' and 'poor' health status according to their baseline ECOG performance or OHA. We hypothesised that patients in 'good' health (0–1 in ECOG classification or "excellent"/"good" in OHA) would report fewer health problems, higher Y-3L index and EQ VAS scores, and lower Y-5L LSS compared to patients who were in poor health (2–4 in ECOG classification or "fair"/"poor"/"very poor" in OHA) [30]. We compared the proportion of reporting any problems in each of the five EQ-5D-Y dimensions between the groups. We used the Chinese Y-3L value set [31] to calculate the Y-3L index scores which range from -0.088 to 1, with higher values indicating better health utility. We calculated the LSS for Y-5L and used it as a proxy of the Y-5L index score since no value set was available for Y-5L. The LSS ranges from 5 (full health) to 25 (the worst possible health). We compared the mean Y-3L, Y-5L, and VAS scores between patients in 'good' and 'poor' health and used Cohen's effect size (ES, difference of mean/pooled SD) to assess the relative efficiency in discriminating between patients in 'good' and 'poor' health.

Responsiveness was assessed using data collected from patients whose baseline and follow-up assessments were made by the same caregiver and whose health status changed from baseline to follow-up according to ECOG, OHA, or treatment status. Specifically, we compared baseline and follow-up Y-3L index, Y-5L LSS and EQ VAS mean scores of patients whose ECOG grade or OHA rating changed to assess responsiveness to altered health status, with a one-category shift in rating indicating either improvement or worsening. Additionally, we hypothesized that chemotherapy would lead to side effects and compared the baseline and follow-up EQ-5D-Y scores of patients who initiated chemotherapy after the baseline assessment but before the follow-up assessment, for assessing responsiveness to deteriorated health. We calculated the standardised effect size (SES) (difference of mean/baseline SD) to evaluate the responsiveness of the EQ measures. SES of 0.20–0.49, 0.5–0.79 and > 0.8 were considered to represent modest, moderate, and good responsiveness.

All analyses were conducted using the statistical package SPSS (IBM SPSS Statistics, Version 26.0, IBM Corp).

Results

A total of 96 paediatric inpatients and their caregivers completed the baseline interviews without any missing responses. The majority of the patients were males (64.6%), primary schoolers (79.2%), with a diagnosis of acute lymphoblastic leukaemia (47.9%); the majority of the caregivers were mothers (67.7%) with the highest education attainment of secondary school or below (66.6%) and residents of urban area (53.1%). The full characteristics of the study participants are displayed in Table 1. The proportion of patients reporting "full health" (i.e. no problems in any health dimension) was 21.9% for Y-3L and 20.8% for Y-5L; the proportion of caregivers reporting "full health" for their child was 24% for Y-3L and 21.9% for Y-5L. The full distributions of patients' and their caregivers' responses to the Y-3L and Y-5L are displayed in Table 2. Eighty-four dyads (87.5%) completed the follow-up interviews with the mean (SD) duration being 2.8 (1.4) days (range: 2 to 13 days). At intervals of 2, 3, and 4 days, there were 25% (21), 44% (37), and 2.4% (2) of dyads, respectively; 12 dyads did not complete the follow-up interviews because of early discharge from hospital ($n = 8$) or caregivers' declination ($n = 4$). There were no significant differences in the characteristics of the dyads who did and did not complete the follow-up interviews except that the children (mean age: 9.3 vs 10.7; $p = 0.044$) and caregivers (mean age: 47.7 vs 39.0; $p = 0.002$) in dyads who did not complete the follow-up interviews were younger and older.

Table 3 shows that higher proportion of patients in 'good' health reported no problems with both Y-3L and Y-5L versions compared to those in 'poor' health, though some of those differences were not statistically significant. For example, 77.1% of 70 patients whose ECOG status was 0 or 1 reported no problems in walking about when completing the Y-3L; in contrast, the proportion of reporting no problems in walking about was only 34.6% based on the responses of the 26 patients whose ECOG status was 2 to

Table 1 Characteristics of patients and caregivers at baseline ($n = 96$)

Patients, % (n) or stated differently		Caregivers, % (*n*) or stated differently	
Age (year), mean (SD)	10.5 (2.2)	Age (year), mean (SD)	40.1 (9.3)
Sex		Relationship	
Male	64.6 (62)	Father	20.8 (20)
Female	35.4 (34)	Mother	67.7 (65)
Educational level		Other	11.5 (11)
Primary school	79.2 (76)	Educational level	
Middle school	20.8 (20)	Primary school or below	20.8 (20)
Disease duration (months), mean (SD)	14.6 (18.8)	Middle school	45.8 (44)
ECOG (grade)		College or above	33.3 (32)
0	16.7 (16)	Residential area	
1	56.3 (54)	Urban	36.4 (35)
2	20.8 (20)	Town	16.7 (16)
3	6.3 (6)	Rural area	44.8 (43)
4	0.0 (0)	Refuse to answer	2.1 (2)
OHA		Monthly personal income	
Excellent	21.8 (21)	< 5000 CNY	40.6 (39)
Good	35.4 (34)	5000–10000 CNY	16.7 (16)
Fair	41.7 (40)	10,001–30000 CNY	12.5 (12)
Poor	1.0 (1)	> 30,000 CNY	6.3 (6)
Very poor	0.0 (0)	Refuse to answer	23.9 (23)
Diagnosis		Marriage	
Acute lymphoblastic leukaemia	47.9 (46)	Unmarried	3.1 (3)
Non-Hodgkin's lymphoma	26.0 (25)	Married	95.8 (92)
Acute myeloid leukaemia	11.5 (11)	Refuse to answer	1.1 (1)
Rhabdomyosarcoma	4.2 (4)	OHA	
Osteosarcoma	3.1 (3)	Excellent	13.5 (13)
Other haematological malignancies	7.3(7)	Good	37.5 (36)
Initial chemotherapy before baseline		Fair	41.7 (40)
Yes	68.8 (66)	Poor	6.3 (6)
No	31.2 (30)	Very poor	1.0 (1)
EQ VAS, mean (SD)	85.8 (15.1)	EQ VAS, mean (SD)	81.2 (14.1)
Y-3L EQ index, mean (SD)	0.85 (0.19)	Y-3L EQ index, mean (SD)	0.84 (0.23)

ECOG Eastern Cooperative Oncology Group performance scale, *OHA* Overall Health Assessment

Table 2 Responses to EQ-5D-Y-3L and EQ-5D-Y-5L descriptive systems by patients and their proxies at baseline ($n=96$)

	Self-reported				Proxy-reported			
	Y-3L		Y-5L		Y-3L		Y-5L	
Dimension	Level	%	Level	%	Level	%	Level	%
Mobility/Walking about	1	65.6	1	60.4	1	74.0	1	67.7
	2	33.3	2	30.2	2	20.8	2	22.9
	3	1.0	3	6.3	3	5.2	3	4.2
			4	2.1			4	2.1
			5	1.0			5	3.1
Looking after myself	1	51.0	1	52.1	1	62.5	1	54.2
	2	41.7	2	30.2	2	32.3	2	32.3
	3	7.3	3	10.4	3	5.2	3	8.3
			4	0.0			4	2.1
			5	7.3			5	3.1
Usual activities	1	61.4	1	59.4	1	69.8	1	70.8
	2	34.4	2	24.0	2	25.0	2	14.6
	3	4.2	3	11.4	3	5.2	3	7.3
			4	1.0			4	3.1
			5	4.2			5	4.2
Pain/discomfort	1	56.3	1	54.2	1	54.2	1	56.2
	2	42.7	2	32.3	2	45.8	2	36.4
	3	1.0	3	11.4	3	0.0	3	6.2
			4	2.1			4	1.0
			5	0.0			5	0.0
Feeling worried /sad/unhappy	1	60.4	1	60.4	1	53.1	1	51.0
	2	38.5	2	27.1	2	41.7	2	34.4
	3	1.0	3	10.4	3	5.2	3	9.4
			4	2.1			4	5.2
			5	0.0			5	0.0

4 to the Y-3L ($p<0.001$, Chi-square test). Similar differences were observed in responses to the Y-3L and Y-5L proxy versions (Table 3). For example, 69.4% of 49 patients whose overall health was rated as 'excellent' or 'good' by their caregivers had no pain or discomfort according to their caregivers' responses to the Y-5L, while only 42.6% of the 47 patients whose overall health was rated as 'fair', 'poor', or 'very poor' experienced no pain or discomfort according to the Y-5L responses of their caregivers ($p=0.007$, Chi-square test).

Table 4 shows the mean Y-3L index, Y-5L LSS and VAS scores for patients in 'good' and 'poor' health and ES values for the differences between the 'good' and 'poor' groups. All the scores differed as expected, that is, compared to patients in 'poor' health, patients in 'good' health had higher Y-3L index and VAS scores and lower Y-5L LSS. Generally, the ES values for the differences in scores based on proxy versions (range: 0.50–2.05) were higher than those based on self-complete versions (range: 0.32–1.38); the ES values for the Y-5L LSS (range: 0.50–2.05) were higher than those for the Y-3L index score (range: 0.32–1.64) (Table 4).

Table 5 shows the mean baseline and follow-up Y-3L index, Y-5L LSS and VAS scores for patients whose health deteriorated and the SES values for the changes in those scores. All the EQ-5D-Y scores worsened, though some of the changes were not statistically significant. For example, the mean (SD) baseline and follow-up Y-3L index scores of patients whose ECOG status worsened ($n=25$) were 0.84 (0.21) and 0.81 (0.18) ($p=0.381$), respectively. The SES values for the score changes based on the self-complete (range: 0.13–0.78) and proxy versions (range: 0.10–0.96) were generally similar; the SES values for the changes in Y-5L LSS (range: 0.31–0.96) were generally higher than the changes in the Y-3L index scores (range: 0.10–0.78) and VAS scores (range: 0.15–0.78).

Table 6 shows the Y-3L index, Y-5L LSS and VAS scores for patients whose health improved and the SES values for the changes in the scores. Although most of the EQ-5D-Y scores improved, most of the changes was not statistically significant and their corresponding SES values were less than 0.2. For example, the baseline and follow-up mean (SD) Y-3L index scores of 12 patients

Table 3 Proportions of patients being reported to have "no problems" by self or a proxy on EQ-5D dimensions in subgroups known to differ in ECOG and overall health status at baseline

EQ-5D-Y version	Known groups	N	Mobility/walking about		Looking after myself		Usual activity		Pain/discomfort		Feeling worried/sad/unhappy	
			%	*p* value	%	*p* value	%	*p* value	%	*p* value	%	*p* value
Y-3L self-complete version												
	ECOG (0–1)	70	77.1	< 0.001	62.9	< 0.001	70.0	0.005	68.6	< 0.001	65.7	0.067
	ECOG (2–4)	26	34.6		19.2		38.5		23.1		46.2	
	OHA (E/G)	55	67.3	0.429	63.6	0.004	67.3	0.126	63.6	0.069	72.7	0.004
	OHA (F/P/V)	41	63.4		34.1		53.7		46.3		43.9	
Y-5L self-complete version												
	ECOG (0–1)	70	77.1	< 0.001	61.4	0.003	67.1	0.011	65.7	< 0.001	65.7	0.067
	ECOG (2–4)	26	15.4		26.9		38.5		23.1		46.2	
	OHA (E/G)	55	67.3	0.084	58.2	0.119	63.6	0.219	63.6	0.025	72.7	0.004
	OHA (F/P/V)	41	51.2		43.9		53.7		41.5		43.9	
Y-3L proxy version												
	ECOG (0–1)	70	90.0	< 0.001	71.4	0.003	78.6	0.003	68.6	< 0.001	61.4	0.007
	ECOG (2–4)	26	30.8		38.5		46.2		15.4		30.8	
	OHA (E/G)	49	83.7	0.023	67.3	0.215	77.6	0.071	73.5	< 0.001	65.3	0.012
	OHA (F/P/V)	47	63.8		57.4		61.7		34.0		40.4	
Y-5L proxy version												
	ECOG (0–1)	70	84.3	< 0.001	62.9	0.005	80.0	0.001	71.4	< 0.001	60.0	0.004
	ECOG (2–4)	26	23.1		30.8		46.2		15.4		26.9	
	OHA (E/G)	49	79.6	0.010	57.1	0.347	75.5	0.211	69.4	0.007	65.3	0.004
	OHA (F/P/V)	47	55.3		51.1		66.0		42.6		36.2	

ECOG Eastern Cooperative Oncology Group performance scale, *OHA* Overall Health Assessment, *E* Excellent, *G* Good, *F* Fair, *P* Poor, *V* Very poor

Table 4 Mean (SD) EQ-5D-Y scores for patients in different health status defined by ECOG and OHA at baseline ($n = 96$)

	ECOG defined subgroups				OHA defined subgroups			
	Grade 0/1 ($n = 70$)	Grade 2–4 ($n = 26$)	*p* value	ES	Excellent/good ($n = 55$, 49) [a]	Fair/poor/very poor ($n = 41$, 47)[b]	*p* value	ES
Self-complete version								
Y-3L index	0.90 (0.14)	0.73 (0.24)	< 0.001	0.99	0.88 (0.19)	0.82 (0.18)	0.122	0.32
Y-5L LSS	7.23 (2.21)	10.65 (3.12)	< 0.001	1.38	7.51 (2.75)	9.02 (2.92)	0.011	0.53
EQ VAS	88.9 (10.7)	77.3 (21.2)	0.001	0.81	90.5 (9.8)	79.4 (18.5)	0.001	0.78
Proxy version								
Y-3L index	0.92 (0.08)	0.62 (0.33)	< 0.001	1.64	0.90 (0.17)	0.77 (0.26)	0.005	0.59
Y-5L LSS	6.63 (1.57)	11.46 (3.75)	< 0.001	2.05	7.18 (2.92)	8.72 (3.29)	0.017	0.50
EQ VAS	84.2 (12.7)	72.9 (14.7)	< 0.001	0.85	87.2 (10.3)	74.8 (14.9)	< 0.001	0.97

a The number of patients reporting "excellent" or "good" OHA was 55 for self-complete version and 49 for proxy version

b The number of patients reporting "fair" or "poor" or "very poor" OHA was 41 for self-complete version and 47 for proxy version

whose OHA improved were 0.79 (0.29) and 0.80 (0.25), respectively ($p = 0.935$; SES = 0.03). With only one exception, the SES values for EQ-5D-Y scores based on the proxy versions were higher than those based on the corresponding self-complete versions. For example, the SES values for the change in the Y-3L index scores of the 17 patients whose ECOG status improved were 0.34 and 0.03 based on the proxy and self-complete versions, respectively.

Table 5 Mean (SD) baseline and follow-up EQ-5D-Y scores for patients whose health status worsened

EQ-5D-Y version and measure	Worsened in ECOG				Worsened in OHA				Chemotherapy initiated			
	Baseline	Follow-up	*p* value	SES	Baseline	Follow-up	*p* value	SES	Baseline	Follow-up	*p* value	SES
Self-complete	($n=25$)				($n=18$)				($n=25$)			
Y-3L index	0.84 (0.21)	0.81 (0.18)	0.381	0.14	0.86 (0.17)	0.79 (0.19)	0.037	0.41	0.85 (0.24)	0.82 (0.18)	0.463	0.13
Y-5L LSS	8.28 (3.53)	10.00 (4.20)	<0.001	0.49	8.22 (3.47)	10.61 (4.49)	<0.001	0.69	7.84 (2.88)	9.28 (3.55)	0.010	0.50
EQ VAS	87.0 (12.7)	79.3 (14.8)	0.009	0.61	87.9 (8.4)	79.7 (16.5)	0.016	0.78	87.8 (10.9)	84.5 (12.7)	0.312	0.30
Proxy version	($n=25$)				($n=11$)				($n=25$)			
Y-3L index	0.86 (0.23)	0.68 (0.35)	0.004	0.78	0.88 (0.10)	0.87 (0.11)	0.458	0.10	0.86 (0.26)	0.80 (0.25)	0.116	0.23
Y-5L LSS	7.80 (3.13)	10.80 (4.19)	<0.001	0.96	7.18 (2.09)	7.82 (2.04)	0.089	0.31	7.44 (3.11)	9.04 (3.22)	0.002	0.51
EQ VAS	81.7 (13.8)	76.2 (14.2)	0.032	0.40	85.0 (10.2)	80.0 (13.2)	0.274	0.49	84.0 (12.9)	82.1 (12.8)	0.429	0.15

Table 6 Mean (SD) baseline and follow-up EQ-5D-Y scores for patients whose health status improved

EQ-5D-Y version and measure	Improved in ECOG				Improved in OHA			
	Baseline	Follow-up	*p* value	SES	Baseline	Follow-up	*p* value	SES
Self-complete	($n=17$)				($n=12$)			
Y-3L index	0.74 (0.29)	0.73 (0.26)	0.970	0.03	0.79 (0.29)	0.80 (0.25)	0.935	0.03
Y-5L LSS	9.88 (3.55)	9.94 (4.16)	0.915	0.02	9.17 (2.79)	8.08 (2.97)	0.090	0.39
EQ VAS	84.9 (12.6)	87.1 (13.0)	0.444	0.17	82.9 (14.2)	87.8 (11.4)	0.263	0.35
Proxy version	($n=17$)				($n=14$)			
Y-3L index	0.70 (0.35)	0.82 (0.18)	0.127	0.34	0.87 (0.09)	0.88 (0.08)	0.857	0.11
Y-5L LSS	10.29 (4.52)	9.00 (3.43)	0.014	0.29	7.14 (1.17)	7.50 (1.70)	0.373	0.25
EQ VAS	81.2 (14.1)	82.4 (11.5)	0.668	0.09	70.9 (17.8)	80.0 (12.2)	0.046	0.51

Discussion

The EQ-5D-Y instruments have demonstrated good measurement properties in some paediatric populations [9, 11, 13, 15, 17]. However, little evidence is available about the performance of the instruments in children and adolescents with haematological malignancies such as leukaemia. In a prior study involving similar patients, we found that both the self-complete and proxy versions of the Y-3L and Y-5L have good test–retest reliability and patient–proxy agreement in children and adolescents with haematological malignancies [32]. To the best of our knowledge, that was the only study of the measurement properties of EQ-5D-Y in this patient population. Other EQ-5D-Y validation studies included children and adolescents with haematological malignancies only as a subgroup of their study samples [16, 24, 33]. In this current study, we simultaneously assessed the validity and responsiveness of the self-complete and proxy versions of the Y-3L and Y-5L in Chinese children and adolescents with haematological malignancies.

Our study provided clear evidence on the validity and responsiveness of the self-complete version of both Y-3L and Y-5L. Specifically, the hypothesis-testing results supported construct validity of all the EQ-5D-Y measures including the items, index, and VAS, and the comparison results of the baseline and follow-up scores indicated responsiveness of the measures to worsened health. To the best of our knowledge, this is the very first evidence for the responsiveness of EQ-5D-Y to worsened health status. The evidence on responsiveness to improvement, however, is not positive. This result could be purely due to chance because it was based on very small numbers of patients. In addition, it is possible that the true improvement the patients experienced was very marginal. Since both baseline and follow-up assessments were conducted during hospital stay when patients waiting for or receiving chemotherapy, clinically meaningful improvement in patients' overall health should be small, if any. This is different from the design of a previous study in which the responsiveness of EQ-5D-Y to improvement was assessed by following up patients 6 to 8 weeks after they received treatment [16, 24]. Therefore, it could be argued that our study design is not suitable for assessing responsiveness to improvement. We anyhow performed this analysis and reported the result in this manuscript for the purpose of transparency. It should be noted that responsiveness to deterioration does not equate to responsiveness to improvement. This is particularly true for EQ-5D instruments whose high ceiling effects are a concern for their ability to capture improved health outcomes. Future studies using better design are needed to further assess the responsiveness of EQ-5D-Y in this patient group.

Our study also supported the construct validity and responsiveness (to deterioration) of the proxy versions of both Y-3L and Y-5L. To date, studies on the validity or responsiveness of the proxy versions of EQ-5D-Y are scarce [34]. Evidence on construct validity was reported mainly for Y-3L [22, 35, 36]; there are only two construct validity studies of Y-5L [24, 36] and one responsiveness study of Y-3L and Y-5L [24]. It should be noted that our effect size results suggested that the proxy version of the two instruments could be more sensitive than the self-complete version to difference and improvement in health outcomes, particularly those clinically defined. To the best of our knowledge, this is the first head-to-head comparison of the validity and responsiveness of the self-complete and proxy versions of the EQ-5D-Y questionnaires. This differential performance between the two versions could be due to the better ability of the caregivers, who are adult, than the patients, who are children or adolescents, to comprehend and respond to the EQ-5D-Y questionnaires. However, there have been few studies that compare the validity and responsiveness of self-complete and proxy versions, particularly in a head-to-head manner.

The effect size results also suggested that, for both self-complete and proxy versions, Y-5L could be more sensitive than Y-3L to difference and improvement in patients' health status, though the two EQ-5D-Y instruments exhibited similar ceiling effects in this study. Head-to-head comparisons of the self-complete version of Y-3L and Y-5L showed mixed results. On one hand, evidence on better construct validity of Y-5L over Y-3L was observed in patients with osteogenesis imperfecta [37], idiopathic scoliosis (for proxy version) [38], and short-sightedness [18], and better responsiveness was observed in patients with haematological diseases (for both self-complete and proxy versions) [16, 24] and traumatic or chronic congenital/acquired orthopaedic condition [19]. On the other hand, Y-5L was not found to be superior to Y-3L in construct validity in a child sample from South Africa [29] or in responsiveness in patients with idiopathic scoliosis [39]. One possible explanation of these mixed findings is that the superiority of Y-5L over Y-3L is disease or sample dependent. That is, the more ill the study subjects, the more salient the value of the Y-5L. If, for example, most of the study subjects experience only slight to moderate health problems in the EQ-5D-Y dimensions, the advantage of Y-5L over Y-3L would be minimal since only the first three levels of the Y-5L are used. It is also possible that the advantage of Y-5L over Y-3L is age dependent, namely, younger children are less able to perceive the difference between the EQ-5D-Y descriptors. As a result, the Y-5L outperforms Y-3L only in children who reach a certain age. Empirical research is warranted to ascertain the reasons for the mixed results on the relative merits of the two EQ-5D-Y instruments. In the meantime, since current evidence suggests that Y-5L is not inferior to Y-3L, it may be a better choice than Y-3L when EQ-5D data is needed for children or adolescents.

This study had several limitations. First, patients and caregivers were not separated when they completed the questionnaires. It is possible that some of them consulted with each other when they answered the EQ-5D questionnaires, although the interviewers instructed them not to do so. Second, our study sample did not allow adequate assessment of EQ-5D-Y's responsiveness to improvement because of very short follow-up periods. Moreover, the responsiveness to chemotherapy initiation was assessed with a very small subgroup of patients. Third, there were no EQ-5D-Y-5L value sets at the time of this study. We used a level sum score as a proxy of the preference-based index score for few patients. The degree to which measurement properties of the level sum score can be generalised to the preference-based index score is unknown for EQ-5D-Y. Nevertheless, a previous study found that the sensitivity of the level sum score and preference-based index score based on EQ-5D-3L and EQ-5D-5L is similar [6]. Lastly, it is worth noting the scarcity of evidence regarding the validation of the OHA proxy version.

In conclusion, this research provided first evidence for the validity and responsiveness of EQ-5D-Y for assessing the health-related quality of life of children and adolescents with haematological malignancies. The proxy and 5-level versions of the instrument demonstrated slightly better sensitivity compared to the self-complete and 3-level versions in this patient group, particularly with respect to known-groups validity and responsiveness to health deterioration.

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Author contributions NL, ZY, PW, BW, MH and JB conceived the study, participated in the design, and made substantial contribution to the intellectual content of the manuscript. WZ, AS and NL participated in the acquisition and interpretation of data, performed statistical analyses and writing of the manuscript. The first draft of the manuscript was written by WZ and AS and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript and agreed to be accountable for all aspects of the work.

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Code availability Not applicable.

Declarations

Competing interests All authors declare that they have no conflicts of interest.

Consent for publication All of the authors read and approved to publish this article. No contents of this article have been published elsewhere.

Ethics approval and consent to participate This study was reviewed and approved by the Ethics Committee of Shanghai Jiaotong University School of Medicine, Renji Hospital Ethics Committee (Project Identification Code: 2018087). The study protocol followed the tenets of the declaration of Helsinki. The participants were fully informed and informed consent was acquired in writing from all of the participating individuals.

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